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Programming Fundamentals

Page & Co. Book Shop Management System

Open Ended Report (SE Semester 1)

Introduction

Problem Statement

The Page and Co. Book Shop Management System is designed to automate and streamline the operations of a bookshop. The system allows users to read books, issue books, return books, order beverages, and generate invoices. The program aims to provide a user-friendly interface for customers while incorporating essential C++ programming concepts to ensure a robust and efficient solution.

Purpose of the Program

The primary purpose of the program is to manage the operations of a bookshop, including book issuance, reading, and returning, as well as handling beverage orders and generating invoices. The program also provides features like currency conversion and premium user discounts to enhance user experience.

Approach Used in Implementation

Core C++ Concepts Implemented

- **Structures**

  The program uses structures to define the Book data type, which includes attributes such as title, author, published Year, and availability. This allows for efficient storage and manipulation of book data.

- **Control Structures**

  The program extensively uses control structures, including loops (for, while) and decision-making statements (if-else, switch), to handle user inputs and control the flow of the program.

  1. Loops: Used to iterate over arrays of books and display them.
  2. Switch Statements: Used to handle user choices in the main menu and within specific functions like issueBook() and readBooks ().

- **Functions**

  The program is modularized using functions to improve readability and reusability. Key functions include:

    a. applyDiscount(double amount): Applies a 10% discount for premium users.

b. convertCurrency(double amount, string currency): Converts the total amount to different currencies.
c. issueBook(): Handles the issuance of books.
d. returnBook(): Manages the return of books and applies late fees if necessary.
e. readBooks(): Displays available books and allows users to read them.
f. orderBeverage(): Handles beverage orders.
g. generateInvoice(): Generates and saves an invoice.
h. showMenu(): Displays the main menu and handles user interactions

- **Data Handling**

Arrays are used to store and manipulate data related to books. Two arrays, fictionBooks and nonFictionBooks, store the details of available books in each genre.

```
Book fictionBooks[10] = {
    {"Notes from the Underground", "Fyodor Dostoevsky", 1864,true},
    {"To Kill a Mockingbird", "Harper Lee", 1960,true},
    {"1984", "George Orwell", 1949,true},
```

- **Input Validation**

The program includes basic input validation to ensure that user inputs are handled correctly. Ex., when a user selects a book to issue or read, the program checks if the selected book number is within the valid range.

User Interface and Usability

The program features a text-based user interface that is easy to navigate. The main menu provides clear options for the user to choose from, and the program provides feedback on user actions, such as successful book issuance or beverage ordering.

Error Handling

The program includes basic error handling to manage invalid user inputs. For example, if a user enters an invalid choice in the main menu, the program displays an error message and prompts the user to try again.

Code Readability and Structure

The code is well-structured and modular, with each function handling a specific task. The use of meaningful variable names and comments enhances code readability. The program is divided into logical sections, making it easy to understand and maintain.

Key Features

Feature	Description
Premium Discount	10% discount applied automatically for premium users.
Book Issuance	Browse fiction/non-fiction books with color-coded availability.
Beverage Ordering	Add tea/coffee/donuts to orders.
Currency Conversion	Convert total to USD, EUR, or PKR.
Late Fee Calculation	Rs 100 penalty for returns after 7 days.
Invoice Generation	Saves invoice to invoice.txt with user details and converted currency.

Challenges Encountered and Solutions Applied

Challenge: Managing Book Availability

One of the challenges was managing the availability of books. The solution was to include an availability attribute in the Book structure and update it accordingly when a book is issued or returned.

Challenge: Currency Conversion

Implementing currency conversion required handling different exchange rates. The solution was to create a convertCurrency() function that takes the amount and the target currency as inputs and returns the converted amount.

Challenge: Generating and Saving Invoices

Generating and saving invoices required handling file I/O operations. The solution was to use the ofstream class to write the invoice details to a text file.

Conclusion

The Page and Co. Book Shop Management System successfully incorporates core C++ concepts such as structures, control structures, functions, and data handling. The program provides a user-friendly interface and includes basic input validation and error handling. The modular design and clear structure of the code make it easy to extend and maintain.

Limitations & Future Enhancements

Limitation	Future Enhancement
Hardcoded book data	Load books from a CSV/JSON file.
Static currency rates	Fetch real-time rates via API.
No user authentication	Add password-based login system.

Submission

The source code for the Page and Co. Book Shop Management System is available on GitHub. The repository includes the main C++ file and a README file with instructions on how to compile and run the program.

GitHub Repository: https://github.com/haseeb-sheikh1/CS1101-Lab-Tasks